# Linear Algebra

Danny Nygård Hansen

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# Preface

These notes cover aspects of linear algebra that I have not found satisfactory expositions of elsewhere. We generally restrict ourselves to the finite-dimensional case, unless results can be generalised without significant effort. For instance, in the context of inner product spaces there is of course no loss in generality by restricting to the real or the complex numbers, and the elementary theory of Hilbert space adjoints is not simplified substantially by the assumption of finite dimension, so we make no such assumption. On the other hand, we only prove the spectral theorem for normal operators on finite-dimensional spaces.

Throughout we let $\mathbb{F}$ denote an arbitrary field, $\mathbb{K}$ a field that is either the real or the complex numbers, and $R$ a commutative ring. Unless otherwise specified, vector spaces will be vector spaces over $\mathbb{F}$, and modules will be left modules over $R$. Furthermore, sesquilinear forms are linear in their *second* entry. This is rarely relevant, but it seems more natural both in the theory of duality of spaces equipped with sesquilinear forms (see par:Riesz-map TODO) and in the representation of sesquilinear forms with matrices (see par:sesquilinear-form-matrix-representation TODO).

# Linear equations and matrices 1

## 1.1 ◇ Linear equations

Let $m$ and $n$ be positive integers. A **linear equation in $n$ unknowns** is an equation on the form

$$l\colon a_1x_1 + \cdots + a_nx_n = b,$$

where $a_1,\ldots,a_n,b \in \mathbb{F}$. A **solution** to $l$ is an element $v = (v_1,\ldots,v_n) \in \mathbb{F}^n$ such that

$$a_1v_1 + \cdots + a_nv_n = b.$$

A **system of linear equations in $n$ unknowns** is a tuple $L = (l_1,\ldots,l_m)$, where each $l_i$ is a linear equation in $n$ unknowns. An element $v \in \mathbb{F}^n$ is a **solution** to $L$ if it is a solution to each linear equation $l_1,\ldots,l_m$.

Let $L$ and $L'$ be systems of linear equations in $n$ unknowns. We say that $L$ and $L'$ are **solution equivalent** if they have the same solutions. Furthermore, we say that they are **combination equivalent** if each equation in $L'$ is a linear combination of the equations in $L$, and vice versa. Clearly, if $L$ and $L'$ are combination equivalent they are also solution equivalent, but the converse does not hold.

## 1.2 ◇ Matrices

For $m,n \in \mathbb{N}$ we denote by $\mathrm{M}_{m,n}(R)$ the set of $m\times n$ matrices over $R$. In the case where $R = \mathbb{F}$, it is well-known that a system of linear equations is equivalent to a matrix equation on the form $Ax = b$, where $A \in \mathrm{M}_{m,n}(\mathbb{F})$, $x \in \mathbb{F}^n$ and $b \in \mathbb{F}^m$. Recall the **elementary row operations** on $A$:

1. multiplication of one row of $A$ by a nonzero scalar,
2. addition to one row of $A$ a scalar multiple of another (different) row, and
3. interchange of two rows of $A$.

If $e$ is an elementary row operation, we write $e(A)$ for the matrix obtained when applying $e$ to $A$. Clearly each elementary row operation $e$ has an 'inverse', i.e. an elementary row operation $e'$ such that $e'(e(A)) = e(e'(A)) = A$. Two matrices $A,B \in \mathrm{M}_{m,n}(\mathbb{F})$ are called **row-equivalent** if $A$ is obtained by applying a finite sequence of elementary row operations to $B$ (and vice versa, though this need not be assumed since each elementary row operation has an inverse).

Clearly, if $A,B \in \mathrm{M}_{m,n}(\mathbb{F})$ are row-equivalent, then the systems of equations $Ax = 0$ and $Bx = 0$ are combination equivalent, hence have the same solutions.

An **elementary matrix** is a matrix obtained by applying a single elementary row operation to the identity matrix $I$. It is easy to show that if $e$ is an elementary row operation and $E = e(I) \in \mathrm{M}_m(\mathbb{F})$, then $e(A) = EA$ for $A \in \mathrm{M}_{m,n}(\mathbb{F})$. If also $B \in \mathrm{M}_{m,n}(\mathbb{F})$, then $A$ and $B$ are row-equivalent if and only if $A = PB$, where $P \in \mathrm{M}_m(\mathbb{F})$ is a product of elementary matrices.

We now show that every matrix is row-equivalent to a matrix with a particularly simple form:

**1.2.1 • Definition.** A matrix $H \in \mathrm{M}_{m,n}(\mathbb{F})$ is called **row-reduced** if

(a) the first nonzero entry of each nonzero row in $H$ is 1, and

(b) each column of $H$ containing the leading nonzero entry of some row has all its other entries equal 0.

If $H$ is row-reduced, it is called a **row-reduced echelon matrix** if it also has the following properties:

(c) Every row of $H$ only containing zeroes occur below every row which has a nonzero entry, and

(d) if rows $1,\dots,r$ are the nonzero rows of $H$, and if the leading nonzero entry of row $i$ occurs in column $k_i$, then $k_1 < \cdots < k_r$. ▲

**1.2.2 • Proposition.** *Every matrix in* $\mathrm{M}_{m,n}(\mathbb{F})$ *is row-equivalent to a unique row-reduced echelon matrix.*

***Proof.*** The usual Gauss–Jordan elimination algorithm proves existence. If $H, K \in \mathrm{M}_{m,n}(R)$ are row-equivalent row-reduced echelon matrices, we claim that $H = K$. We prove this by induction in $n$. If $n = 1$ then this is obvious, so assume that $n > 1$. Let $H_1$ and $K_1$ be the matrices obtained by deleting the $n$th column in $H$ and $K$ respectively. Then $H_1$ and $K_1$ are also row-equivalent[1] and row-reduced echelon matrices, so by induction $H_1 = K_1$. Thus if $H$ and $K$ differ, they must differ in the $n$th column.

Let $H_2$ be the matrix obtained by deleting columns in $H$, only keeping those columns containing pivots, as well as keeping the $n$th column. Define $K_2$ similarly. Thus we have deleted the same columns in $H$ and $K$, so $H_2$ and $K_2$ are also row-equivalent. Say that the number of columns in $H_2$ and $K_2$ is $r+1$, and write the matrices on the form

$$H_2 = \begin{pmatrix} I_r & h \\ 0 & h' \end{pmatrix} \quad \text{and} \quad K_2 = \begin{pmatrix} I_r & k \\ 0 & k' \end{pmatrix},$$

[1] It should be obvious that deleting columns preserves row-equivalence, but we give a more precise argument: If $P \in \mathrm{M}_m(\mathbb{F})$ is a product of elementary matrices and $a_1,\dots,a_n \in \mathbb{F}^m$ are the columns in $A$, then the columns in $PA$ are $Pa_1,\dots,Pa_m$. Thus elementary row operations are applied to each column independently of the other columns.

where $h, k \in \mathbb{F}^r$ and $h', k' \in \mathbb{F}^{m-r}$ are column vectors. Since $H_2$ and $K_2$ are row-equivalent, the systems $H_2 x = 0$ and $K_2 x = 0$ are solution equivalent. If $h' = 0$, then $H_2 x = 0$ has the solution $(-h, 1)$. But this is also a solution to $K_2 x = 0$, so $h = k$ and $k' = 0$. If $h' \neq 0$, then $H_2 x = 0$ only has the trivial solution. But then $K_2 x = 0$ also only has the trivial solution, and hence $k' \neq 0$. But that must be because both $H_2$ and $K_2$ has a pivot in the rightmost column, so also in this case $H_2 = K_2$. ∎

Next we study when square matrices are invertible. The main result says for a square matrix to be invertible, it suffices that it has either a left- or a right-inverse. Since (as we will see in par:matrix-rep TODO) square matrices are precisely the linear endomorphisms on finite-dimensional vector spaces, this shows that for such an endomorphism to be invertible, it suffices that it has either a left- or a right-inverse. This result is also a direct consequence of the rank–nullity theorem, see e.g. Roman (2008, Corollary 2.9) (TODO and local ref).

Notice that elementary matrices are invertible since elementary row operations are invertible.

**1.2.3 • Lemma.** *If $A \in \mathrm{M}_n(\mathbb{F})$, then the following are equivalent:*

(i) *$A$ is invertible,*

(ii) *$A$ is row-equivalent to $I_n$,*

(iii) *$A$ is a product of elementary matrices, and*

(iv) *the system $Ax = 0$ has only the trivial solution $x = 0$.*

***Proof.*** *(i) ⇔ (ii)*: Let $H \in \mathrm{M}_n(\mathbb{F})$ be a row-reduced echelon matrix that is row-equivalent to $A$. Then $H = PA$, where $P \in \mathrm{M}_n(\mathbb{F})$ is a product of elementary matrices. Then $A = P^{-1}H$, so $A$ is invertible if and only if $H$ is. But the only invertible row-reduced echelon matrix is the identity matrix, so (i) and (ii) are equivalent.

*(ii) ⇒ (iii)*: As above, there exists a product $P$ of elementary matrices such that $I_n = PA$, so $A = P^{-1}$.

*(iii) ⇒ (i)*: This is obvious since elementary matrices are invertible.

*(ii) ⇔ (iv)*: If $A$ and $I_n$ are row-equivalent, then the systems $Ax = 0$ and $I_n x = 0$ have the same solutions. Conversely, assume that $Ax = 0$ only has the trivial solution. If $H \in \mathrm{M}_{m,n}(\mathbb{F})$ is a row-reduced echelon matrix that is row-equivalent to $A$, then $Hx = 0$ has no nontrivial solution. Thus if $r$ is the number of nonzero rows in $H$, then $r \geq n$. But then $r = n$, so $H$ must be the identity matrix. ∎

**1.2.4 • Proposition.** *Let $A \in \mathrm{M}_n(\mathbb{F})$. Then the following are equivalent:*

(i) *$A$ is invertible,*

(ii) *$A$ has a left inverse, and*

(iii) *A has a right inverse.*

***Proof.*** If $A$ has a left inverse, then $Ax = 0$ has no nontrivial solution, so $A$ is invertible. If $A$ has a right inverse $B \in \mathrm{M}_n(\mathbb{F})$, i.e. $AB = I$, then $B$ has a left inverse and is thus invertible. But then $A$ is the inverse of $B$ and hence is itself invertible. ∎

# Bases and coordinates 2

## 2.1 ⋄ Bases

If $\mathcal{V}$ is a subset of $V$, the **span** of $\mathcal{V}$, denoted $\operatorname{span}\mathcal{V}$ or $\langle\mathcal{V}\rangle$, is the smallest subspace of $V$ containing $\mathcal{V}$. Equivalently, it is the set of all linear combinations

$$\alpha_1 v_1 + \cdots + \alpha_n v_n,$$

where $\alpha_i \in \mathbb{F}$ and $v_i \in \mathcal{V}$. We say that $\mathcal{V}$ is **linearly independent** if any linear relation

$$\alpha_1 v_1 + \cdots + \alpha_n v_n = 0$$

among elements $v_i$ in $\mathcal{V}$ can only be satisfied if $\alpha_1 = \cdots = \alpha_n = 0$. If $\mathcal{V}$ is both linearly independent and a spanning set, then we call it a **basis** for $V$. We have the following characterisation of bases:

**2.1.1 • Proposition.** *A subset $\mathcal{V} \subseteq V$ is a basis for $V$ if and only if*

$$V = \bigoplus_{v\in\mathcal{V}} \langle v\rangle. \qquad \blacksquare$$

We next prove that bases always exist, but first we need a different characterisation of bases. An element $v \in V$ is an **essentially unique** linear combination of the elements in $\mathcal{V}$ if there is an up to ordering unique way to express $v$ as a linear combination of elements in $\mathcal{V}$. It is easy to see that $\mathcal{V}$ is linearly independent if and only if every nonzero $v \in \langle\mathcal{V}\rangle$ is an essentially unique linear combination of the elements in $\mathcal{V}$.

**2.1.2 • Proposition.** *Let $\mathcal{V}$ be a subset of $V$. The following are equivalent:*

(i) *$\mathcal{V}$ is linearly independent and spans $V$.*

(ii) *Every nonzero $v \in V$ is an essentially unique linear combination of vectors in $\mathcal{V}$.*

(iii) *$\mathcal{V}$ is a minimal spanning set.*

(iv) *$\mathcal{V}$ is a maximal linearly independent set.*

***Proof.*** *(i) ⇔ (ii)*: This follows easily as mentioned above.

*(i) ⇔ (iii)*: If (i) holds and a proper subset $\mathcal{V}'$ of $\mathcal{V}$ spanned $V$, then any element of $\mathcal{V} \setminus \mathcal{V}'$ is a linear combination of elements in $\mathcal{V}'$, so $\mathcal{V}$ is not linearly independent. Conversely, if $\mathcal{V}$ is a minimal spanning set but is not linearly independent, then some $v \in \mathcal{V}$ is a linear combination of the other elements in $\mathcal{V}$, so $\mathcal{V} \setminus \{v\}$ is also a spanning set.

*(i) ⇔ (iv)*: Again assuming (i), if $\mathcal{V}$ were not maximal there would be some $v \in V \setminus \mathcal{V}$ such that $\mathcal{V} \cup \{v\}$ were linearly independent. But then $v$ would not be a linear combination of elements in $\mathcal{V}$. Conversely, if $\mathcal{V}$ is a maximal linearly independent set that did not span $V$, then there would be some $v \in V \setminus \mathcal{V}$ that is not a linear combination of elements in $\mathcal{V}$. But then $\mathcal{V} \cup \{v\}$ is also linearly independent. ∎

**2.1.3 • Theorem.** *Let $V$ be a vector space. If $\mathcal{I} \subseteq V$ is linearly independent, $\mathcal{S} \subseteq V$ is a spanning set, and $\mathcal{I} \subseteq \mathcal{S}$, then there is a basis $\mathcal{V}$ for $V$ with $\mathcal{I} \subseteq \mathcal{V} \subseteq \mathcal{S}$.*

***Proof.*** Let $\mathcal{A}$ be the collection of linearly independent subsets $\mathcal{J}$ of $V$ with $\mathcal{I} \subseteq \mathcal{J} \subseteq \mathcal{S}$. If $\mathcal{C}$ is a chain in $\mathcal{A}$, then

$$\mathcal{U} = \bigcup \mathcal{C}$$

is linearly independent and satisfies $\mathcal{I} \subseteq \mathcal{U} \subseteq \mathcal{S}$, so it lies in $\mathcal{A}$. Hence every chain in $\mathcal{A}$ has an upper bound, so it has a maximal element $\mathcal{V}$. This is linearly independent since it lies in $\mathcal{A}$, and it is also a spanning set by maximality, hence it is a basis. ∎

**2.1.4 • Corollary.** *Every vector space has a basis.*

***Proof.*** Let $\mathcal{I} = \emptyset$ and $\mathcal{S} = V$ in Theorem 2.1.3. ∎

We next turn to the concept of the **dimension** of a vector space. Our presentation will focus on finite-dimensional vector spaces.

**2.1.5 • Proposition.** *If the vectors $v_1, \dots, v_n$ in $V$ are linearly independent, and the vectors $w_1, \dots, w_m$ span $V$, then $n \leq m$.*

***Proof.*** List the vectors as follows:

$$w_1, \dots, w_m; v_1, \dots, v_n.$$

We transform this list such that the collection of vectors on the left-hand side of the semicolon always span $V$, and such that the vectors on the right-hand side are always linearly independent. Note that $v_1$ is a linear combination of the $w_j$, implying that we may add $v_1$ to the left-hand side and remove one of the $w_j$ (which, by reindexing, we may assume is $w_1$) and still have a spanning set. We simultaneously remove $v_1$ from the right-hand side. That is, we obtain

$$v_1, w_2, \dots, w_m; v_2, \dots, v_n.$$

If $m < n$, then applying this process recursively will eventually exhaust the $w_j$, at which point we would have

$$v_1, \dots, v_m; v_{m+1}, \dots, v_n.$$

But this is not possible, since $v_n$ does not lie in the span of $v_1, \dots, v_m$. Hence $n \leq m$. ∎

**2.1.6 • Corollary.** *If $V$ has a finite spanning set, then all bases for $V$ have the same cardinality.* ■

This in fact holds for arbitrary vector spaces, though the proof is significantly more involved (cf. Roman 2008, Theorem 1.12).

Since bases always exist and all bases have the same cardinality, the following definition makes sense:

**2.1.7 • Definition: *Dimension.***
The **dimension** of a vector space $V$, written $\dim V$, is the cardinality of any basis for $V$. ▲

We now turn to a different characterisation of the dimension of finite-dimensional vector spaces. Below we write $\dim V = \infty$ if the dimension of the vector space $V$ is infinite. A **series** of subspaces $U_i$ of $V$ is a finite or infinite decreasing sequence

$$V = U_0 \supsetneq U_1 \supsetneq U_2 \supsetneq \cdots.$$

If the sequence is finite, then the **length** of the series is the number of strict inclusions. If the sequence is infinite, then we say that the length of the series is $\infty$. The maximal length of a series of subspaces of $V$ is denoted $l(V)$.

In the proposition below, we write $\dim V = \infty$ if the dimension of $V$ is infinite.

**2.1.8 • Proposition.** *Let $V$ be a vector space. Then* $\dim V = l(V)$.

***Proof.*** First assume that $V$ is finite-dimensional, and let $\mathcal{V} = (v_1, \ldots, v_n)$ be a basis for $V$. Then there is a series

$$V = \langle v_1, \ldots, v_n \rangle \supsetneq \langle v_1, \ldots, v_{n-1} \rangle \supsetneq \cdots \supsetneq \langle v_1 \rangle \supsetneq 0$$

of subspaces of $V$, so $\dim V \leq l(V)$. Conversely, let

$$V = U_0 \supsetneq U_1 \supsetneq U_2 \supsetneq \cdots$$

be a series of subspaces of $V$. If the series ends with $0$, remove it. Hence all subspaces in the series are nontrivial. Then choose for each $i$ an element $v_i \in U_i \setminus U_{i+1}$, and collect them in a set $\mathcal{I}$. It is clear that $\mathcal{I}$ is linearly independent, hence finite. Thus the series is also finite with length $|\mathcal{I}| - 1$. Adding back $0$ to the series we obtain a series that is at least as long as the original sequence, and that is of length $|\mathcal{I}| \leq \dim V$. Since the sequence was arbitrary, $l(V) \leq \dim V$.

Next assume that $V$ is infinite-dimensional. Then $V$ contains a sequence $(v_i)_{i \in \mathbb{N}}$ that is linearly independent, so the series

$$V \supseteq \langle v_i \mid i \in \mathbb{N} \rangle \supsetneq \langle v_i \mid i \geq 2 \rangle \supsetneq \langle v_i \mid i \geq 3 \rangle \supsetneq \cdots$$

is infinite, and $l(V) = \infty$. Conversely, assume that $V$ has an infinite series. As above we construct a linearly independent set $\mathcal{I}$ whose size equals the length of the sequence. Thus $V$ contains an infinite linearly independent set, so $\dim V = \infty$. ■

## 2.2 ◇ Coordinate maps and matrices

Every matrix $A \in \mathrm{M}_{m,n}(\mathbb{F})$ gives rise to a map $M_A \colon \mathbb{F}^n \to \mathbb{F}^m$ given by $M_A v = Av$. The next result shows that every linear map $\mathbb{F}^n \to \mathbb{F}^m$ arises in this way:

**2.2.1 • Proposition.** *Let $(e_1, \dots, e_n)$ be the standard basis for $\mathbb{F}^n$. The map*

$$\mathcal{M} \colon \mathcal{L}(\mathbb{F}^n, \mathbb{F}^m) \to \mathrm{M}_{m,n}(\mathbb{F}), \qquad T \mapsto \big(Te_1 \mid \cdots \mid Te_n\big),$$

*is a linear isomorphism with inverse $A \mapsto M_A$. The matrix $\mathcal{M}(T)$ is called the* ***standard matrix representation*** *of $T$. If $T \colon \mathbb{F}^n \to \mathbb{F}^m$ and $S \colon \mathbb{F}^m \to \mathbb{F}^l$ are linear maps, then*

(i) *$Tv = \mathcal{M}(T)v$ for all $v \in \mathbb{F}^n$.*

(ii) *$\mathcal{M}(\mathrm{id}_{\mathbb{F}^n}) = I$.*

(iii) *$\mathcal{M}(S \circ T) = \mathcal{M}(S)\mathcal{M}(T)$.*

(iv) *$T$ is invertible if and only if $\mathcal{M}(T)$ is invertible, in which case $\mathcal{M}(T^{-1}) = \mathcal{M}(T)^{-1}$.*

***Proof.*** The map $A \mapsto M_A$ is clearly linear, so to prove the first point it suffices to show that this is the inverse of $\mathcal{M}$. Let $T \in \mathcal{L}(\mathbb{F}^n, \mathbb{F}^m)$. Then

$$M_{\mathcal{M}(T)} \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{pmatrix} = \mathcal{M}(T) \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{pmatrix} = \big(Te_1 \mid \cdots \mid Te_n\big) \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{pmatrix} = \sum_{i=1}^n \alpha_i Te_i = T \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{pmatrix}$$

for $\alpha_1, \dots, \alpha_n \in \mathbb{F}$. Conversely, for $A \in \mathrm{M}_{m,n}(\mathbb{F})$ we have

$$\mathcal{M}(M_A) = \big(M_A e_1 \mid \cdots \mid M_A e_n\big) = \big(Ae_1 \mid \cdots \mid Ae_n\big) = A,$$

since $Ae_i$ is the $i$th column of $A$. We prove the remaining claims:

*(i)*: Simply notice that $Tv = M_{\mathcal{M}(T)}v = \mathcal{M}(T)v$.

*(ii)*: This is obvious from the definition of $\mathcal{M}$.

*(iii)*: Let $v \in \mathbb{F}^n$ and notice that

$$\mathcal{M}(S \circ T)v = (S \circ T)v = S(Tv) = S(\mathcal{M}(T)v) = \mathcal{M}(S)\mathcal{M}(T)v$$

by (i). Since this holds for all $v$, the claim follows.

*(iv)*: This follows easily from (ii) and (iii). ■

Having characterised all linear maps between powers of the field $\mathbb{F}$, we now wish to characterise the linear maps between abstract finite-dimensional vector spaces. The first order of business is to establish a correspondence between such a vector space and an appropriate power of $\mathbb{F}$.

Let $V$ be a finite-dimensional $\mathbb{F}$-vector space. If $\mathcal{V} = (v_1, \dots, v_n)$ is an ordered basis for $V$, then for every $v \in V$ there are unique $\alpha_1, \dots, \alpha_n \in \mathbb{F}$ such that $v = \sum_{i=1}^n \alpha_i v_i$. Hence the map $\varphi_{\mathcal{V}} \colon V \to \mathbb{F}^n$ given by $\varphi_{\mathcal{V}}(v) = (\alpha_1, \dots, \alpha_n)$ is well-defined. Furthermore, it is clearly linear, and since $\mathcal{V}$ is a basis it is also bijective, hence a linear isomorphism. The map $\varphi_{\mathcal{V}}$ is called the **coordinate map** with respect to $\mathcal{V}$, and the vector $[v]_{\mathcal{V}} = \varphi_{\mathcal{V}}(v)$ is called the **coordinate vector** of $v$ with respect to $\mathcal{V}$.

Now let $\mathcal{W}$ be another ordered basis for $V$. The composition $\varphi_{\mathcal{W},\mathcal{V}} = \varphi_{\mathcal{W}} \circ \varphi_{\mathcal{V}}^{-1}$ is called the **change of basis operator** from $\mathcal{V}$ to $\mathcal{W}$, and this makes the diagram

$$\begin{tikzcd} & \mathbb{F}^n \arrow[dd, "\varphi_{\mathcal{W},\mathcal{V}}"] \\ V \arrow[ur, "\varphi_{\mathcal{V}}"] \arrow[dr, "\varphi_{\mathcal{W}}"'] & \\ & \mathbb{F}^n \end{tikzcd} \tag{2.1}$$

commute. Its standard matrix is denoted ${}_{\mathcal{W}}[\square]_{\mathcal{V}}$. This has the expected properties:

**2.2.2 • Proposition.** *Let $\mathcal{V}$, $\mathcal{W}$ and $\mathcal{U}$ be ordered bases for a finite-dimensional $\mathbb{F}$-vector space $V$. Then*

(i) $[v]_{\mathcal{W}} = \varphi_{\mathcal{W},\mathcal{V}}([v]_{\mathcal{V}})$ *for all $v \in V$. In particular,* $[v]_{\mathcal{W}} = {}_{\mathcal{W}}[\square]_{\mathcal{V}} \cdot [v]_{\mathcal{V}}$.

(ii) $\varphi_{\mathcal{V},\mathcal{V}}$ *is the identity map. In particular,* ${}_{\mathcal{V}}[\square]_{\mathcal{V}}$ *is the identity matrix.*

(iii) $\varphi_{\mathcal{U},\mathcal{W}} \circ \varphi_{\mathcal{W},\mathcal{V}} = \varphi_{\mathcal{U},\mathcal{V}}$. *In particular,* ${}_{\mathcal{U}}[\square]_{\mathcal{W}} \cdot {}_{\mathcal{W}}[\square]_{\mathcal{V}} = {}_{\mathcal{U}}[\square]_{\mathcal{V}}$.

(iv) $\varphi_{\mathcal{W},\mathcal{V}}$ *(resp.* ${}_{\mathcal{W}}[\square]_{\mathcal{V}}$*) is invertible with inverse* $\varphi_{\mathcal{V},\mathcal{W}}$ *(resp.* ${}_{\mathcal{V}}[\square]_{\mathcal{W}}$*).*

***Proof.*** All claims about change of basis matrices follow by Proposition 2.2.1 from the corresponding claims about change of basis operators.

The claim (i) follows by commutativity of the diagram (2.1), i.e.

$$\varphi_{\mathcal{W},\mathcal{V}}([v]_{\mathcal{V}}) = (\varphi_{\mathcal{W}} \circ \varphi_{\mathcal{V}}^{-1}) \circ \varphi_{\mathcal{V}}(v) = \varphi_{\mathcal{W}}(v) = [v]_{\mathcal{W}}.$$

Claim (ii) is an immediate consequence of the definition of $\varphi_{\mathcal{V},\mathcal{V}}$. The remaining claims are proved similarly to (i). ∎

Next consider a linear map $T \colon V \to W$. If $\mathcal{V} \in V^n$ and $\mathcal{W} \in W^m$ are bases for $V$ and $W$ respectively, then the diagram

$$\begin{tikzcd} V \arrow[r, "\varphi_{\mathcal{V}}"] \arrow[d, "T"'] & \mathbb{F}^n \arrow[d, "\varphi_{\mathcal{W}} \circ T \circ \varphi_{\mathcal{V}}^{-1}"] \\ W \arrow[r, "\varphi_{\mathcal{W}}"'] & \mathbb{F}^m \end{tikzcd}$$

commutes. The map $\varphi_{\mathcal{W}} \circ T \circ \varphi_{\mathcal{V}}^{-1}$ is the **basis representation** of $T$ with respect to the bases $\mathcal{V}$ and $\mathcal{W}$. This is a linear map $\mathbb{F}^n \to \mathbb{F}^m$, so it has a standard matrix which we denote ${}_{\mathcal{W}}[T]_{\mathcal{V}}$. This is called the **matrix representation** of $T$ with respect to the bases $\mathcal{V}$ and $\mathcal{W}$.

**2.2.3 • Proposition.** *Let $V$ and $W$ be finite-dimensional $\mathbb{F}$-vector spaces with ordered bases $\mathcal{V} = (v_1, \dots, v_n) \in V^n$ and $\mathcal{W} \in W^m$, respectively. The map*

$$\begin{aligned} {}_{\mathcal{W}}[\,\cdot\,]_{\mathcal{V}} \colon \mathcal{L}(V, W) &\to \mathrm{M}_{m,n}(\mathbb{F}), \\ T &\mapsto {}_{\mathcal{W}}[T]_{\mathcal{V}}, \end{aligned}$$

*is a linear isomorphism. Let $T \colon V \to W$ and $S \colon W \to U$ be linear maps, and let $\mathcal{U} \in U^l$ be an ordered basis for $U$. Then*

(i) ${}_{\mathcal{W}}[T]_{\mathcal{V}} = \big([Tv_1]_{\mathcal{W}} \mid \cdots \mid [Tv_n]_{\mathcal{W}}\big)$.

(ii) $[Tv]_{\mathcal{W}} = {}_{\mathcal{W}}[T]_{\mathcal{V}} \cdot [v]_{\mathcal{V}}$ *for all* $v \in V$.

(iii) *If $\mathcal{V}'$ is another basis for $V$, then* ${}_{\mathcal{V}'}[\mathrm{id}_V]_{\mathcal{V}} = {}_{\mathcal{V}'}[\square]_{\mathcal{V}}$.

(iv) ${}_{\mathcal{U}}[S \circ T]_{\mathcal{V}} = {}_{\mathcal{U}}[S]_{\mathcal{W}} \cdot {}_{\mathcal{W}}[T]_{\mathcal{V}}$.

(v) *$T$ is invertible if and only if ${}_{\mathcal{W}}[T]_{\mathcal{V}}$ is invertible, in which case* ${}_{\mathcal{V}}[T^{-1}]_{\mathcal{W}} = {}_{\mathcal{W}}[T]_{\mathcal{V}}^{-1}$.

***Proof.*** For the first claim, notice that the map $T \mapsto \varphi_{\mathcal{W}} \circ T \circ \varphi_{\mathcal{V}}^{-1}$ is a linear isomorphism, since pre- and postcomposition with linear isomorphisms are themselves linear isomorphisms. Composing this map with $\mathcal{M}$ yields ${}_{\mathcal{W}}[\,\cdot\,]_{\mathcal{V}}$, so this is a linear isomorphism by Proposition 2.2.1.

*(i)*: If $(e_1, \dots, e_n)$ is the standard basis for $\mathbb{F}^n$, then the definition of the standard matrix representation yields that the $i$th column of ${}_{\mathcal{W}}[T]_{\mathcal{V}}$ is given by

$${}_{\mathcal{W}}[T]_{\mathcal{V}} \cdot e_i = \mathcal{M}(\varphi_{\mathcal{W}} \circ T \circ \varphi_{\mathcal{V}}^{-1}) \cdot e_i = (\varphi_{\mathcal{W}} \circ T \circ \varphi_{\mathcal{V}}^{-1}) e_i = \varphi_{\mathcal{W}}(Tv_i) = [Tv_i]_{\mathcal{W}},$$

as claimed.

*(ii)*: Notice that

$$\begin{aligned} [Tv]_{\mathcal{W}} &= (\varphi_{\mathcal{W}} \circ T)(v) \\ &= (\varphi_{\mathcal{W}} \circ T \circ \varphi_{\mathcal{V}}^{-1}) \circ \varphi_{\mathcal{V}}(v) \\ &= (\varphi_{\mathcal{W}} \circ T \circ \varphi_{\mathcal{V}}^{-1})([v]_{\mathcal{V}}) \\ &= {}_{\mathcal{W}}[T]_{\mathcal{V}} \cdot [v]_{\mathcal{V}}. \end{aligned}$$

where the last equality follows from Proposition 2.2.1(i).

*(iii)*: This is obvious from the definitions of ${}_{\mathcal{V}'}[\mathrm{id}_V]_{\mathcal{V}}$ and ${}_{\mathcal{V}'}[\square]_{\mathcal{V}}$.

*(iv)*: Notice that

$$\varphi_{\mathcal{U}} \circ (S \circ T) \circ \varphi_{\mathcal{V}}^{-1} = (\varphi_{\mathcal{U}} \circ S \circ \varphi_{\mathcal{W}}^{-1}) \circ (\varphi_{\mathcal{W}} \circ T \circ \varphi_{\mathcal{V}}^{-1})$$

The claim then follows from Proposition 2.2.1(iii).

*(v)*: This is an immediate consequence of either (iv) or of Proposition 2.2.1(iv). ■

**2.2.4 • Proposition.** *Let $\mathcal{V} = (v_1, \dots, v_n)$ be an ordered basis for an $\mathbb{F}$-vector space $V$, and let $T\colon V \to V$ be a linear isomorphism. Let $\mathcal{W} = (w_1, \dots, w_n)$ where $w_i = Tv_i$. Then $\mathcal{W}$ is an ordered basis for $V$ and*

$$\varphi_{\mathcal{W},\mathcal{V}} = \varphi_{\mathcal{V}} \circ T^{-1} \circ \varphi_{\mathcal{V}}^{-1}, \quad \text{or} \quad {}_{\mathcal{W}}[\square]_{\mathcal{V}} = {}_{\mathcal{V}}[T^{-1}]_{\mathcal{V}}.$$

*In particular, if $V = \mathbb{F}^n$ and $\mathcal{V}$ is the standard basis $\mathcal{E}$, then*

$$\varphi_{\mathcal{W},\mathcal{E}} = T^{-1}, \quad \text{or} \quad {}_{\mathcal{W}}[\square]_{\mathcal{E}} = \mathcal{M}(T)^{-1}.$$

We think of this result as follows: If we change basis by applying an invertible linear transformation $T$, we obtain the coordinate vectors corresponding to the transformed basis by applying $T^{-1}$ (in the old basis). This says that if we perform a *passive transformation*, i.e. a change of basis while keeping vectors themselves fixed, the coordinates change by the inverse of said transformation.

***Proof.*** Let $v \in V$ and write $v = \sum_{i=1}^n \alpha_i v_i$. Then

$$Tv = \sum_{i=1}^n \alpha_i T v_i = \sum_{i=1}^n \alpha_i w_i = \varphi_{\mathcal{W}}^{-1} \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{pmatrix} = \varphi_{\mathcal{W}}^{-1} \circ \varphi_{\mathcal{V}}(v),$$

implying that

$$\varphi_{\mathcal{W},\mathcal{V}} = \varphi_{\mathcal{W}} \circ \varphi_{\mathcal{V}}^{-1} = (T \circ \varphi_{\mathcal{V}}^{-1}) \circ^{-1} \varphi_{\mathcal{V}}^{-1} = \varphi_{\mathcal{V}} \circ T^{-1} \circ \varphi_{\mathcal{V}}^{-1}$$

as claimed. ■

# 3 Structural properties of vector spaces

## 3.1 ◇ Projections I

Let $V$ be a vector space. A linear operator $P\colon V \to V$ is called a **projection** if it is idempotent, i.e., if $P^2 = P$.

**3.1.1 • Proposition.** *A linear map $P\colon V \to V$ is a projection if and only if there exist subspaces $U$ and $W$ of $V$ such that $V = U \oplus W$, $P|_U = \iota_U$ and $P|_W = 0$. In this case $U = \operatorname{im} P$ and $W = \ker P$.*

We say that $P$ is the projection onto $U$ along $W$.

***Proof.*** Assume that $P$ is a projection, and let $v \in \operatorname{im} P$. Then $v = Pu$ for some $u \in V$, and

$$Pv = P^2 u = Pu = v.$$

If also $v \in \ker P$, then $v = 0$. Furthermore, for any $v \in V$ we have $v = Pv + (v - Pv) \in \operatorname{im} P \oplus \ker P$, so $\operatorname{im} P$ and $\ker P$ are indeed complements in $V$.

The converse is obvious, and so is the characterisation of $U$ and $W$. ∎

We will return to projections in sec:projections-2 TODO.

## 3.2 ◇ Quotient spaces and complements

If $U$ is a subspace of an $\mathbb{F}$-vector space $V$, then its underlying additive group is a subgroup of the underlying additive group of $V$. Since $V$ considered as such is abelian, we may consider the quotient group $V/U$ whose elements are cosets $v + U$ for $v \in V$. It is then trivial to check that the operation $\alpha(v + U) := \alpha v + U$ for $\alpha \in \mathbb{F}$ makes $V/U$ into a vector space. We denote by $\pi_U$ or simply by $\pi$ the quotient map $\pi\colon V \to V/U$ given by $\pi(v) = v + U$.

**3.2.1 • Theorem.** *Let $U$ be a subspace of $V$. If $T\colon V \to W$ satisfies $U \subseteq \ker T$, then there is a unique linear map $\tilde{T}\colon V/U \to W$ such that the diagram*

$$\begin{tikzcd} & W \\ V \arrow[ur, "T"] \arrow[dr, "\pi"'] & \\ & V/U \arrow[uu, dashed, "\tilde{T}"'] \end{tikzcd}$$

*commutes.*

***Proof.*** The corresponding result for groups yields a unique group homomorphism $\tilde{T}$. This is easily seen to also be a linear map. ■

This has the following immediate consequence:

**3.2.2 • COROLLARY:** ***Canonical decomposition.***
*Every linear map $T\colon V \to W$ may be decomposed as follows:*

$$\begin{tikzcd} V \arrow[r, two heads, "\pi"'] \arrow[rrr, bend left, "T"] & V/\ker T \arrow[r, "\sim", "\tilde{T}"'] & \operatorname{im} T \arrow[r, hook, "\iota_{\operatorname{im} T}"'] & U \end{tikzcd}$$

*In particular we have the* ***first isomorphism theorem****: $V/\ker T \cong \operatorname{im} T$.* ■

If $U$ is a subspace of $V$, then a subspace $W$ of $V$ with the property that $V = U \oplus W$ is called a **complement** of $U$. Complements are certainly not unique, but we have the following:

**3.2.3 • LEMMA.** *Assume that $V$ has two direct sum compositions*

$$U \oplus W_1 = V = U \oplus W_2,$$

*where $W_1 \subseteq W_2$. Then $W_1 = W_2$.*

***Proof.*** Assume that $v \in W_2$. Then there exist unique $u \in U$ and $w \in W_1$ such that $v = u + w$. But then $w$ also lies in $W_2$, and uniqueness implies that $u = 0$ and $w = v$. But then $v \in W_1$ as desired. ■

Next we note the following characterisation of complements:

**3.2.4 • PROPOSITION.** *Let $U$ be a subspace of $V$, and let $W$ be a complement of $U$. The projection $P$ onto $W$ along $U$ induces an isomorphism $V/U \cong W$.*

***Proof.*** Note that $\ker P = U$ and $\operatorname{im} P = W$ by Proposition 3.1.1, so Corollary 3.2.2 implies that $W \cong V/U$ as claimed. ■

So far in this section we have not made use of the fact that all vector spaces have bases. This fact enters the present discussion through the following result:

**3.2.5 • PROPOSITION.** *Every subspace $U$ of a vector space $V$ has a complement.*

***Proof.*** Choose a basis $\mathcal{U}$ for $U$ and extend it to a basis $\mathcal{V}$ for $V$ using Theorem 2.1.3. Then we clearly have $V = U \oplus \langle \mathcal{V} \setminus \mathcal{U} \rangle$. ■

If $U$ is a subspace of $V$, then the dimension of the quotient space $V/U$ is called the **codimension** of $U$ in $V$ and is denoted $\operatorname{codim}_V U$ or simply $\operatorname{codim} U$. The results above then implies the following:

**3.2.6 • Corollary.** *If $U$ is a subspace of $V$, then*

$$\dim V = \dim U + \operatorname{codim} U. \qquad \blacksquare$$

**3.2.7 • Corollary: *The rank–nullity theorem.***
*Let $T \in \mathcal{L}(V, W)$. Then* $\operatorname{codim} \ker T = \dim \operatorname{im} T$*, and in particular*

$$\dim V = \dim \ker T + \dim \operatorname{im} T. \qquad \blacksquare$$

## 3.3 ◇ Linear maps

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